

Detecting and Preventing Phishing Attacks on Health Workers through Network Traffic Monitoring and Detection

Problem Identification:

Phishing attacks on health workers are increasing, especially during health crises, because attackers exploit urgency and trust to steal credentials, access patient records, or install ransomware.

Key Vulnerability: Health workers often use insecure networks (e.g., public Wi-Fi in field hospitals). Health workers are often not informed on how to detect phishing emails, and health systems may lack phishing detection at the network layer.

Research and Analysis:

Case Study:

As documented by the Health Sector Cybersecurity Coordination Centre (HC3). On Thursday, August 3rd, 2023, a ransomware-as-a-service group called Rhysida attacked a healthcare system that operates 16 hospitals and more than 165 clinics and outpatient centres in four states through a phishing email (ransomware attack).

The attack was successful via phishing, which led to penetration of the network and encryption of the data using a sophisticated encrypted protocol known as a 4096-bit RSA key with the ChaCha20 algorithm. This protocol locks the data and denies access to the systems. The attacker then locks files and folders by calling PowerShell to delete the binary after encryption is completed.

How Rhysida's ransomware works

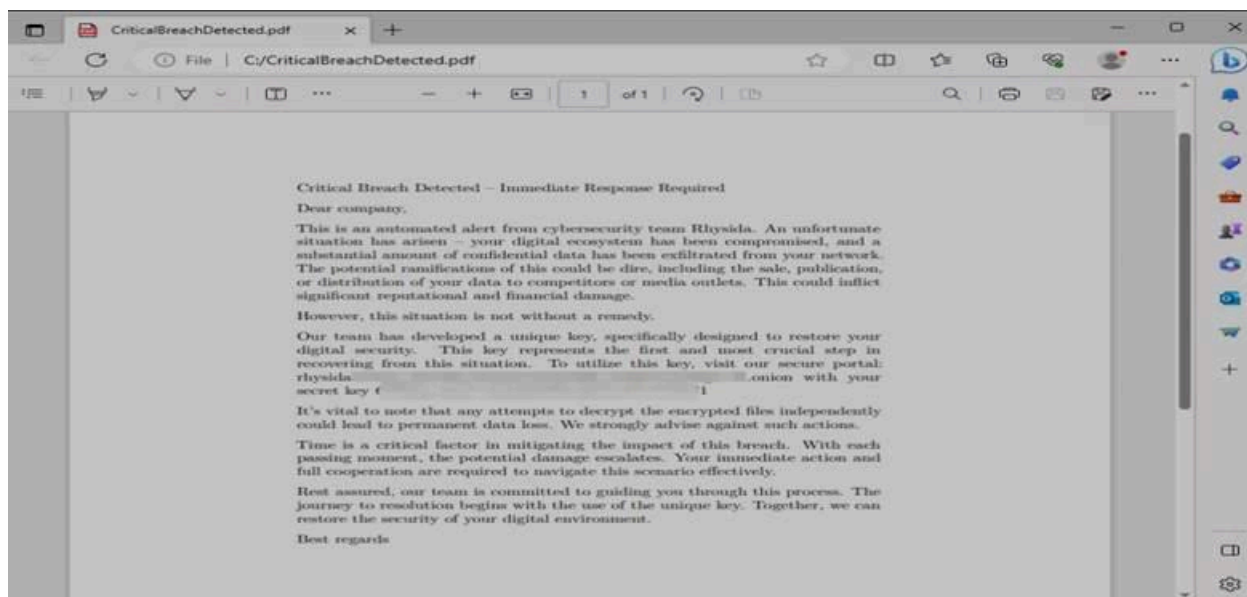
Rhysida's ransomware is a 64-bit Portable Executable (PE) Windows cryptographic ransomware application compiled using MINGW/GCC. According to the researcher, when Rhysida runs, the researcher observed a process of getting output from the command line, which scans the files, runs the "file_to_crypt" function, and if successful, changes the file extension to ".rhysida"

```
Select C:\a864282fea5a536510ae86c77ce46f7827687783628e4f2ceb5bf2c41b8cd3c6a864282fea5a536510ae86c77ce46f782768778...
Current dir entry edb.log
Current dir entry edbres00001.jrs
Current dir entry edbres00002.jrs
Current dir entry edbtmp.log
Current dir entry IndexedDB.edb
Current dir entry IndexedDB.jfm
Current dir entry LocalCache
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AC/Microsoft/Internet Explorer/DOMStore/1E703AEC/windows.msn[1].xml.rhysida
Directory C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/LocalCache entries 0
Current dir entry LocalState
Directory C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/LocalState entries 5
Current dir entry AppIconCache
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AppData/IndexedDB/IndexedDB.edb.rhysida
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AppData/CacheStorage/CacheStorage.edb.rhysida
Directory C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/LocalState/AppIconCache entries 1
Current dir entry 100
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AppData/IndexedDB/edbres00002.jrs.rhysida
Directory C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/LocalState/AppIconCache/100 entries 225
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AppData/IndexedDB/edb.log.rhysida
ERROR open file to_crypt C:/[redacted]/AppData/Local/Packages/Microsoft.Windows.Search_cw5n1h2txyewy/AC/Microsoft/CryptnetUrlCache/MetaData/B398880134F722095474390821A8308D_A4CF52CCA82D7458083F7280801A3A04.rhysida
Current dir entry C:_ProgramData_chocolatey_bin_apimonitor-x64_exe
Current dir entry C:_ProgramData_chocolatey_bin_apimonitor-x86_exe
```

Script outputs that appear on cmd[.]exe when Rhysida runs

☐	platform.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pdb.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pickle.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pickletools.py.rhysida	7/25/2023 2:39 AM	RHYSIDA File
☐	pipes.py.rhysida	7/25/2023 2:39 AM	RHYSIDA File
☐	pkgutil.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	platform.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	plistlib.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	poplib.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	posixpath.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pprint.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	profile.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☒	pstats.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pty.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	py_compile.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	pyclbr.py.rhysida	7/25/2023 2:39 AM	RHYSIDA File
☐	pydoc.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	queue.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	quopri.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	random.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File
☐	re.py.rhysida	7/25/2023 2:42 AM	RHYSIDA File

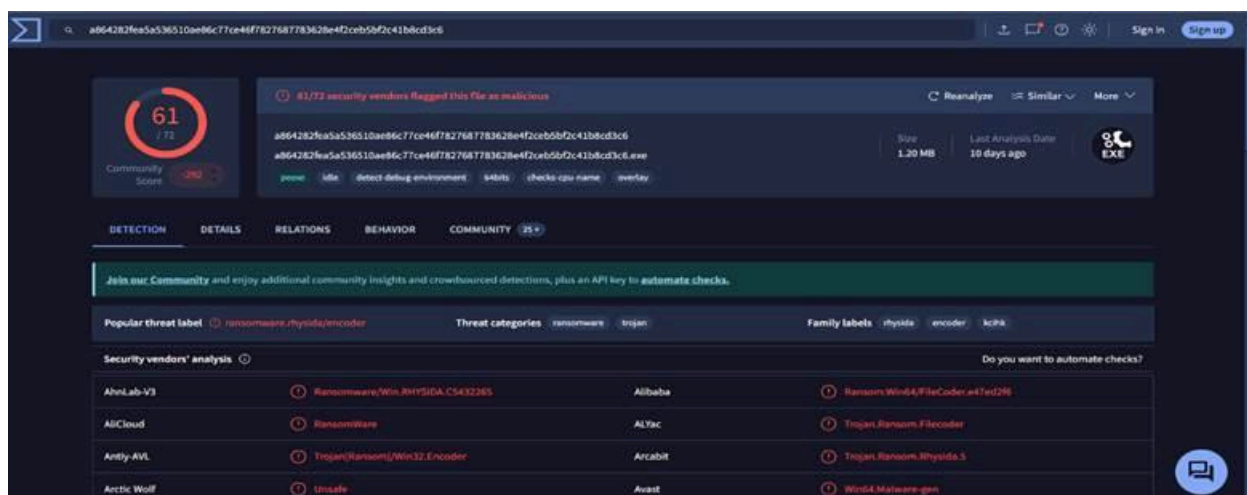
Some files in socradar.io test environment changed to “.rhysida” extension



Rhysida Ransomware's ransom note 'CriticalBreachDetected.pdf'

Analysis of IoCs of Rhysida Ransomware:

We used VirusTotal to analyse one of the file hashes, "Hash (SHA-256) a864282fea5a536510ae86c77ce46f7827687783628e4f2ceb5bf2c41b8cd3c6," which was reported as malicious. Over 61 vendors have reported the hash as malicious. Upon further investigation, this file hash is known as the ransomware.rhysida/encoder, which has been commonly used by the advanced threat actor Rhysida.



Here are other lists of Hashes that have been flagged as malicious:

Hash (SHA-256) 6903b00a15eff9b494947896f222bd5b093a63aa1f340815823645fd57bd61de

Hash (SHA-1) 7abc07e7f56fc27130f84d1c7935a0961bd58cb9

Hash (SHA-256) 3bc0340007f3a9831cb35766f2eb42de81d13aeb99b3a8c07dee0bb8b000cb96

Hash (SHA-256)

2a3942d213548573af8cb07c13547c0d52d1c3d72365276d6623b3951bd6d1b2

Hash (MD-5) 59a9ca795b59161f767b94fc2dece71a

Hash (SHA-256) 250e81eeb4df4649ccb13e271ae3f80d44995b2f8ffca7a2c5e1c738546c2ab1

Impacts of the attack:

As a result of the attack, it forces the emergency department to close, ambulances to be diverted, and other medical services to cease operations, which makes the files, systems and connected medical devices no longer available to the health care system, exposing sensitive patient data (violating HIPAA/GDPR) and leading to loss of public trust and psychological stress for health workers. These types of breaches not only put patient privacy at risk but can also threaten lives when operations are disrupted.

Note: If attackers could penetrate big hospitals and clinics in urban areas using social engineering like phishing emails, which is a **key attack vector**, it would be much easier to penetrate the underserved population if proper measures/solutions are not provided to healthcare workers, especially during health crises, e.g, COVID-19.

Solution Development:

Build a phishing detection system designed specifically to protect healthcare workers in underserved populations from phishing attacks. It combines both signature-based and heuristic detection methods to identify suspicious emails in .txt, .eml, and .html formats. The system offers two user interfaces: a command-line tool (CLI) and an easy-to-use Flask web interface, making it accessible for both tech-savvy and non-technical users. It can scan uploaded email files or connect directly to a live inbox via IMAP using the authenticator app rather than passwords.

When a phishing attempt is detected, the system automatically sends an alert to the healthcare worker, helping prevent data breaches or ransomware incidents that could disrupt critical healthcare services. All detection results are saved as CSV or JSON reports and securely stored in a database, with user authentication to protect access.

How It Helps Underserved Populations:

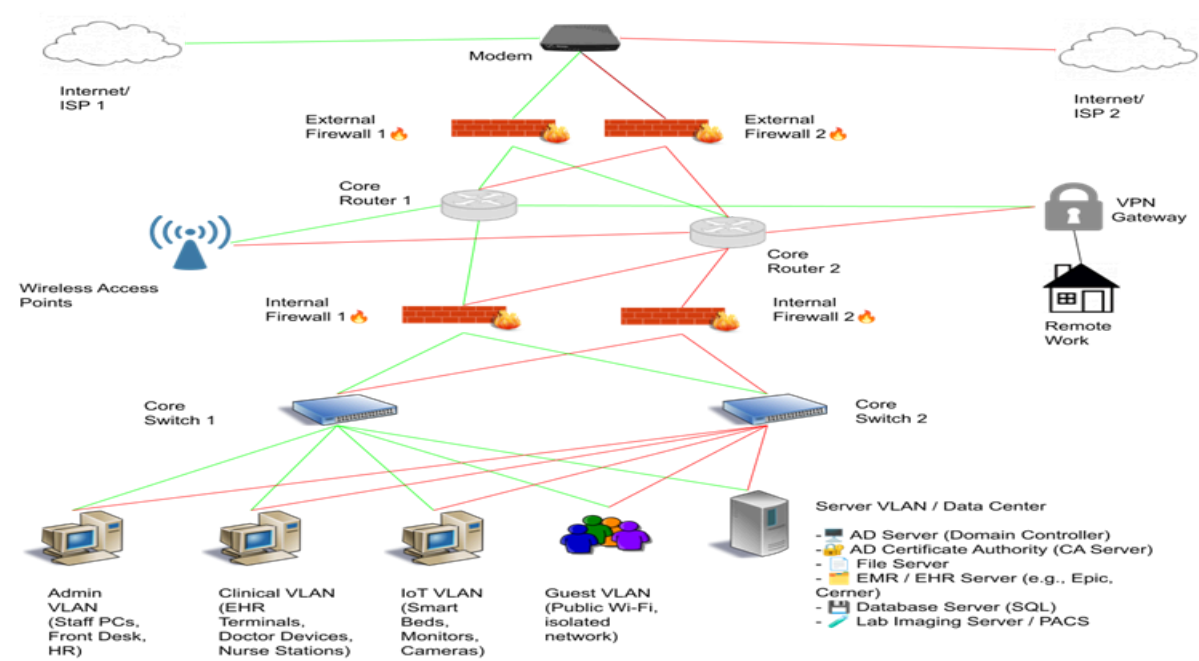
By providing affordable, accessible phishing protection tools, this project empowers healthcare workers in underserved regions who often lack a strong IT security infrastructure. Preventing phishing attacks in these settings helps safeguard patient data, maintain the continuity of healthcare services, and reduce the risk of financial or operational damage to already

resource-limited healthcare providers. This contributes directly to improving healthcare resilience and supports UN SDG Goal 3: Good Health and Well-being.

```
phishguard/
|
|— app.py           # Main Flask app
|— detector.py      # Email analysis engine
|— imap_scanner.py  # Inbox scanning logic
|— alerts.py        # Email alert system
|— vt_scan.py       # VirusTotal integration
|— models.py        # Database models
|— auth.py          # User login routes
|— config.py        # App settings
|— whitelist.json   # Whitelisted domains and keywords
|— templates/       # HTML UI with Tailwind
|— uploads/         # Uploaded files
|— report.csv/json  # Generated reports
|— requirements.txt # Dependencies
|— README.md
```

Project Structure

HC3 Attack Response: Network Schematic for Healthcare Resilience



Network Configuration Timeline and Environment Justification

To strengthen the digital infrastructure of the affected healthcare system following the ransomware attack by the Rhysida group, a secure network environment will be developed to reduce the likelihood of future intrusions, ensure system availability, and protect sensitive patient

data. The network schematic prioritises logical segmentation, layered security, and infrastructure redundancy, which are key principles necessary to support the continuity of operations within healthcare environments that depend on constant system accessibility.

The estimated implementation timeline for this configuration spans approximately 6 to 10 weeks. Initial weeks are focused on establishing VLAN segmentation across departments and device types, followed by the deployment and fine-tuning of firewall rules, VPN access points, and routing logic. Subsequent phases involve hardening the domain and server infrastructure, including Active Directory, certificate authorities, EMR/EHR systems, SQL databases, and lab imaging servers, before introducing redundant routing and connectivity through dual-core switches, routers, and ISPs. The final stages include network-wide testing, failover simulations, and operational readiness checks, which encompass staff training and documentation validation.

This environment creates isolated traffic zones between clinical, administrative, IoT (Internet of Things), guest, and server networks, with multiple internal and external firewalls to limit lateral movement and unauthorised access. The design also includes a dedicated VPN gateway for secure remote access, ensuring connectivity without exposing internal systems to unnecessary external threats. Redundant infrastructure further reinforces availability, reducing the risk of downtime during an attack or system failure. Together, these components establish a defence-in-depth strategy aligned with best practices in the health sector cybersecurity.

Post-Setup Continuity and Maintenance Strategy

A secure network environment must be supported by an ongoing maintenance and continuity strategy to remain effective after deployment. The following measures are essential to uphold network security, system reliability, and regulatory compliance over time:

- **Configuration Backups**
Routine backups of firewalls, switches, routers, VPN configurations, and servers should be performed and securely stored. Backup procedures must include regular restoration tests to validate recovery processes.
- **Patch Management**
Firmware, operating systems, and application software should be updated on a regular schedule. Vulnerability assessments must be conducted routinely to detect and remediate outdated systems or misconfigurations.
- **Log Management and Monitoring**
Centralised logging using a SIEM platform enables early detection of threats, policy violations, and suspicious activity. Logs should be collected from firewalls, VPNs, endpoint systems, and core servers.
- **Redundancy and Failover Testing**
Regular testing of redundant switches, routers, and ISP failover paths ensures business continuity in the event of hardware or service disruptions. Simulated outage scenarios should be used to measure responsiveness.

Implementing Education and Awareness for Phishing Attack Prevention on Health Workers:

As part of strengthening defences against phishing attacks targeting healthcare workers, it's essential to go beyond technical detection tools and invest in continuous education and awareness activities. While our project focuses on detecting phishing attempts through network traffic monitoring, we recognise that technology alone is not enough — the human element plays a critical role in protection.

Under the NIST PR.AT-1 guideline, we recommend implementing an education program that equips healthcare employees and partners with the knowledge and tools to safeguard their organisation. Common phishing schemes they should learn to recognise include:

- Credential theft: Emails trying to trick them into revealing usernames and passwords.
- Malware delivery: Messages prompting them to download malicious attachments or software, often used to spread ransomware.
- Financial scams (BEC): Requests for gift card purchases, fake invoices, or fraudulent wire transfers.

To build a resilient defence, the training program should cover:

- The proper use of encryption tools within email systems reduces the likelihood of sensitive data being intercepted.
- Caution when handling emails with sensitive data (PII or PHI); double-check recipient addresses before sending, and be vigilant about sharing confidential information.
- Clear reporting procedures for suspicious emails - ensuring that all staff know how to quickly alert the IT team or designated security contact, who can then investigate and provide further instructions.

By combining robust technical defences like network traffic monitoring with practical user training, we aim to create a layered, people-centred defence that reduces risk, protects patient data, and ensures uninterrupted healthcare delivery in underserved regions.

Implementation Plan:

Phase	Description	Timeline	Responsibility
Planning	Define system requirements, successful metrics, and gather phishing use cases.	2-3 weeks	Project Manager, IT Security
Development	Build and test CLI + Flask phishing detection (signature + heuristic), IMAP integration.	6-8 weeks	Dev Team, QA Team
Network Setup	Apply VLAN segmentation, firewalls, VPNs, and harden servers; ensure redundancy and patching.	3-4 weeks	Network Engineers, Sys Admins
Integration	Connect the detection system to inboxes, set up alerts, and store results in a database with authentication.	3-4 weeks	Integration Lead, DB Admin
Training	Provide phishing awareness training, distribute materials, and set up reporting processes (NIST PR.AT-1).	2-3 weeks	Training Lead, Security Lead
Testing	Run system + network tests, phishing simulations, and failover drills.	1-2 weeks	QA Team, Incident Response
Go-Live	Deploy the system, enable monitoring, and hand over to operations.	1-2 weeks	Project Manager, Support Lead
Maintenance	Apply regular updates, monitor logs, run backups, and deliver ongoing training.	Continuously	IT Ops, Security Team

Sources

1. Health Sector Cybersecurity Coordination Center (HC3) Health Sector Cybersecurity Coordination Center. "Rhysida Ransomware." U.S. Department of Health and Human Services, 3 Aug. 2023.
<https://www.hhs.gov/sites/default/files/rhysida-ransomware-hc3-analyst-note.pdf>
2. Cybersecurity & Infrastructure Security Agency (CISA) Cybersecurity and Infrastructure Security Agency. "#StopRansomware: Rhysida Ransomware." 15 Nov. 2023.
<https://www.cisa.gov/news-events/cybersecurity-advisories/aa23-319a>
3. Krebs on Security (Expert Commentary) Krebs, Brian. "BlackCat Ransomware Group Claims Change Healthcare Attack." KrebsOnSecurity, 22 Feb. 2024.
<https://krebsonsecurity.com/2024/02/blackcat-ransomware-group-claims-change-healthcare-attack>
4. VirusTotal. "File Hash Lookup and Threat Analysis for a864282fea5a536510ae86c77ce46f7827687783628e4f2ceb5bf2c41b8cd3c6." Google, 2023.
<https://www.virustotal.com/>
5. National Institute of Standards and Technology (NIST). Framework for Improving Critical Infrastructure Cybersecurity. Version 1.1, NIST, 2018.
<https://nvlpubs.nist.gov/nistpubs/CSWP/NIST.CSWP.04162018.pdf>
6. OWASP Foundation. "Phishing Detection." OWASP, 2023.
<https://owasp.org/www-community/Phishing>
7. Common Vulnerabilities and Exposures (CVE) MITRE. "Common Vulnerabilities and Exposures (CVE)." MITRE Corporation, 2023.
<https://cve.mitre.org/>
8. CIS Controls v8 Center for Internet Security. "CIS Critical Security Controls Version 8." 2021.
<https://www.cisecurity.org/controls/cis-controls-list>
9. Home. total assignment help. (n.d.).
<https://www.totalassignment.com/free-sample/wireless-network-design-assignment-project-for-jones-hospital>